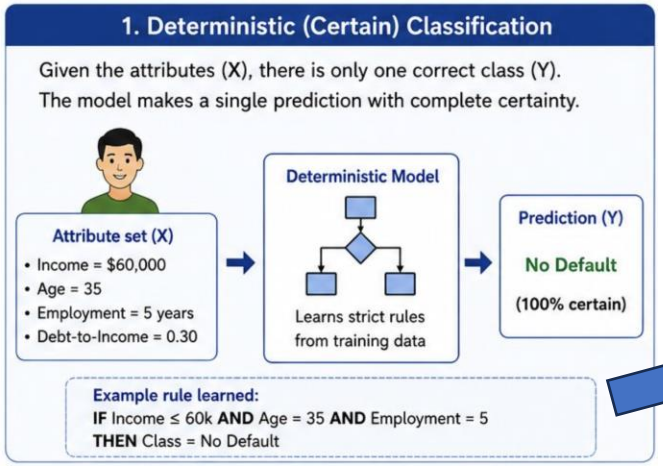


CSE5DMI Data Mining

Week 06
Classification: Alternative Techniques

Deterministic vs Probabilistic

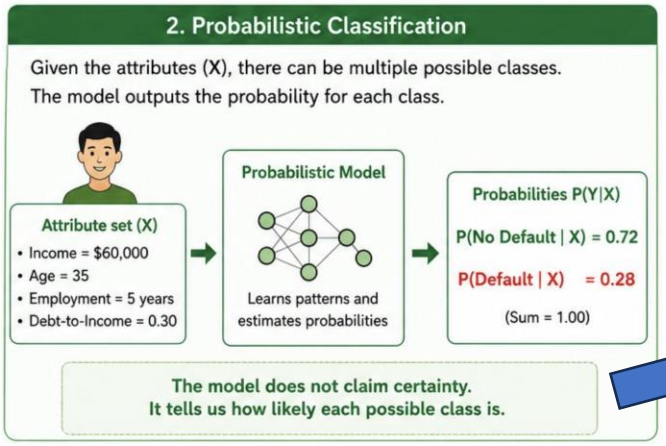
- In the previous topic, we studied **deterministic classifiers**, such as rule-based classifiers.
  - These models make a definite decision: **given a set of attributes X, the model assigns the observation to a single class Y.**



e.g. Rule-based classifier

We say: Given these attributes the class should be Y

- However, in many real-world problems, the same attributes do not always lead to a completely certain outcome.
  - **How likely is this instance to belong to each possible class?"**



e.g. Bayes Classifier

We say: Given these attributes, how likely is the possible class

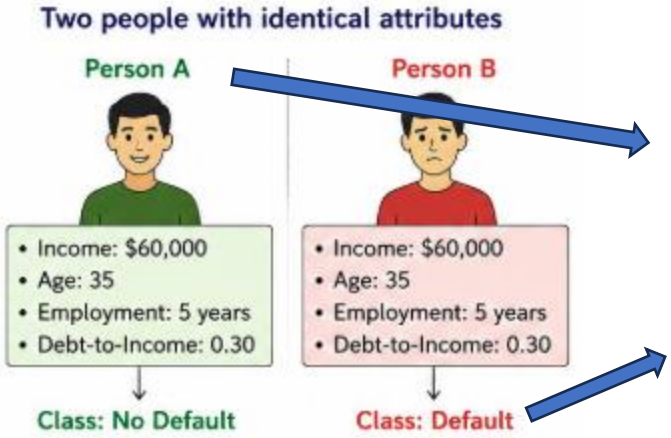
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Bayes Classifier (Illustration)

Example: Loan Default Prediction

Attributes (X)				Class (Y) (Default?)
Income ($)	Age (years)	Employment (Years)	Debt-to-Income Ratio	
60,000	35	5	0.30	No
60,000	35	5	0.30	Yes
...	...	...	...	...
60,000	35	5	0.30	No

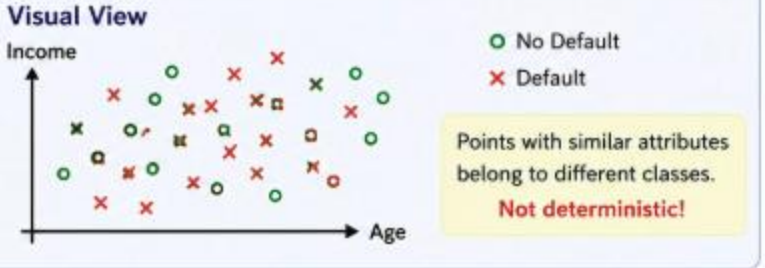


In many real-world problems, the same attributes (same data) do not always lead to a same class (outcome)

Why?

There may be unobserved factors, noise in the data, or other hidden influences (e.g., financial habits, health issues, emergencies) that affect the outcome.



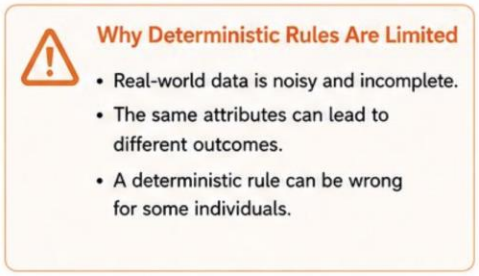


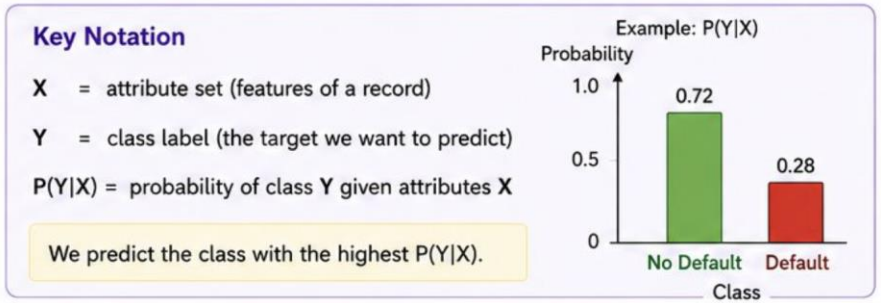
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Bayes Classifier

- Many cases relationship between the attribute set and class labels is non-deterministic
- The class labels of a test record cannot be predicted with certainty even though its attribute set is identical to some of training examples.
- This situation may arise due to noisy data or presence of certain cofounding factors that affect classification. - see previous slide
- Bayesian classifier presents an approach for modelling probabilistic relationship between the attribute set and class variables.





Takeaway: Probabilistic classifiers explicitly handle uncertainty, noise, and missing factors. This makes them **more suitable for many real-world classification problems.**

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Bayes Classifier

- A probabilistic framework for solving classification problems
- Conditional Probability:

$$P(Y \mid X) = \frac{P(X,Y)}{P(X)}$$

"Probability of observing Y conditioned on the outcome of X" or "Given X, what is the probability of Y happening"

$$P(X \mid Y) = \frac{P(X,Y)}{P(Y)}$$

"Probability of observing X conditioned on the outcome of Y" or "Given Y, what is the probability of X happening"

- Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

In mathematics and statistics, it is proved that the conditional probability is related to the joint probability. In this bayes classifier, P(Y|X) is also called posterior probability

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Example of Sam and Mark case

Given:

- **Sam and Mark are friends.**
  - **History information knows that 50% of the time, Sam is going to party given Mark is in the party**
  - **We call it *Likelihood* (P(S|M) = 0.5)**
- **Prior probability of Mark going to party 1/50,000 - (P(M))**
- **Probability that Sam goes to the party: 1/20**
  - **(P(S) -** probability of the observed evidence = 1 / 20)

We called it posterior probability P (M|S) here because we update probability of Mark going after observing the evidence that Sam is going."

If Sam is going to the party, what's the Posterior Probability of Mark going to the party?

$$P(M \mid S) = \frac{P(S \mid M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

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Bayes Classifier

Bayes theorem is useful because it allows us to express posterior probability in terms of prior probability P(Y), *the likelihood or class conditional probability* *P(X | Y)*, and the evidence P(X).

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

The denominator *P(X)* is always constant and can be ignored.

$P(Y)$ can easily be calculated from the training set.

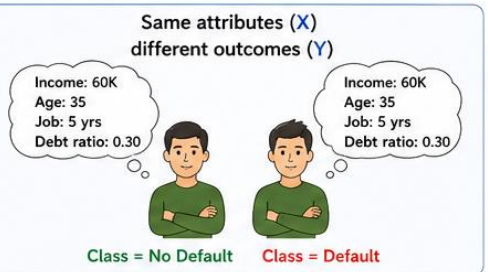
$P(X|Y)$ can be implemented or calculated using Naive Bayes Classifier.

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Bayes Classifier Summary

Bayesian Classification — From Attributes to Posterior Probability

- Let X denote the attribute set and Y denote the class variable.
- Consider each attribute and class label has a nondeterministic relationship, then we can treat them as random variables.
- P(Y | X) captures the relationship probabilistically.
- This conditional probability is also known as the posterior probability as opposed to prior probability P(Y).



Because the relationship is not deterministic, attributes and class are treated as random variables.

1) TRAINING DATA

ID	Attributes (X)				Class (Y) (Default?)
	Income ($K)	Age (years)	Job Tenure (years)	Debt Ratio	
1	55	28	2	0.40	No
2	60	35	5	0.30	No
3	45	40	3	0.60	Yes
4	70	30	6	0.20	No
5	50	36	1	0.70	Yes
...	...	...	...	...	...
N	65	33	4	0.35	No

2) POSTERIOR PROBABILITY P(Y | X)

For a new record X, we update our belief about each class using the attributes. We compute P(Y | X).

Prior P(Y) → Likelihood P(X | Y): How likely are these attributes if the class is Y? → Evidence P(X): Same for all classes → ignored when comparing → Posterior P(Y | X): Updated probability of each class after seeing X



$$P(Y \mid X) = \frac{P(X \mid Y) \times P(Y)}{P(X)} \propto P(X \mid Y) \times P(Y) \quad \text{(compare across classes)}$$

3) DURING TRAINING: LEARN P(Y | X)

The model learns posterior probabilities for many combinations of attribute values and classes from the training data.

Some combinations of X	Learned Posterior Probabilities P(Y | X)	
(Income, Age, Job Tenure, Debt Ratio)	P(No Default | X)	P(Default | X)
(55, 28, 2, 0.40)	0.85	0.15
(60, 35, 5, 0.30)	0.72	0.28
(45, 40, 3, 0.60)	0.20	0.80
...	...	...

→ Model stores many such probabilities

4) CLASSIFY A NEW TEST RECORD X'

New test record X'

Income = 60K
Age = 35
Job Tenure = 5 yrs
Debt Ratio = 0.30

→ Compute posterior probabilities

P(No Default | X') = 0.72

P(Default | X') = 0.28

↓

Choose class with the largest posterior probability

Predicted class Y' = No Default

Class with highest P(Y | X') is predicted.

Key takeaway: Bayesian classification does not give a single certain rule. It estimates how likely each class is for a given set of attributes, then predicts the class with the highest posterior probability.

Notation Summary
X = attribute set
Y = class variable
P(Y | X) = posterior probability

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Case study

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

What is the class label using Bayes Classifier?

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

P (No | X)
P (Yes | X)
Which is greater?

P (No | X) = P (X | No) * P(No)
P (Yes | X) = P(X | Yes) * P (Yes)
Which is greater?

How to estimate probabilities from Data?

Given Test Record $X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

P(X | Yes) =

P(Refund = No | Yes) x

P(Divorced | Yes) x

P(Income = 120K | Yes)

P(X | No) =

P(Refund = No | No) x

P(Divorced | No) x

P(Income = 120K | No)

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How to Estimate Probabilities from Data

- For discrete attributes:
  - Counting

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

P(X | Yes) =

P(Refund = No | Yes) x

P(Divorced | Yes) x

P(Income = 120K | Yes)

P(X | No) =

P(Refund = No | No) x

P(Divorced | No) x

P(Income = 120K | No)

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Another Example – step 1

Bayes' Theorem: $P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$

Training Data

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Summary

1 The Training Data

20 animals, 4 attributes + class.
We want to classify a new animal (A).

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
...	...	...	...	...	...
Total = 20 animals				7 mammals	13 non-mammals

Test Instance or unseen sample

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

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Another Example – step 2

Bayes' Theorem: $P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$

Training Data

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Calculate P(Y) for each class

2 Prior Probabilities $P(Y)$

Count how many in each class.

Class (Y)	Count	Prior Probability $P(Y)$
Y = mammals	7	$P(Y = \text{mammals}) = \frac{7}{20} = 0.35$
Y = non-mammals	13	$P(Y = \text{non-mammals}) = \frac{13}{20} = 0.65$

These are our prior beliefs before seeing the new animal X.

Test Instance or unseen sample

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

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Another Example – step 3

Bayes' Theorem: $P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$

Training Data

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Observe The Test Instance

3 The Test Instance X

We want to classify this new animal with the following features:

Give Birth	Can Fly	Live in Water	Have Legs	Class (Y) ?
yes	no	yes	no	?

We will compute $P(X \mid Y = \text{mammals})$ and $P(X \mid Y = \text{non-mammals})$, multiply by the priors $P(Y)$, and compare.

Notation: Y = mammals, Y = non-mammals, X = the test instance

Test Instance or unseen sample

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

We will compute **conditional probabilities** *P(X|Y=mammals) and P(X|Y=non-mammals) for next steps*

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Another Example – step 4

Bayes' Theorem: $P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$

Training Data

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Observe the conditional probabilities or likelihood the class mammals from 7 instances of training data

4 Likelihood: $P(X \mid Y = \text{mammals})$

Compute the probability of each attribute value given $Y = \text{mammals}$). (There are 7 mammals.)

Attribute	Value in X (x_i)	Count with this value & Y = mammals	Total mammals	$P(x_i \mid Y = \text{mammals})$
Give Birth	yes	6	7	$\frac{6}{7}$
Can Fly	no	6	7	$\frac{6}{7}$
Live in Water	yes	2	7	$\frac{2}{7}$
Have Legs	no	2	7	$\frac{2}{7}$

Multiply (Naïve Bayes assumption):

$$P(X \mid Y = \text{mammals}) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} \approx 0.06$$

Counts come from the training table (7 mammals total).

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Another Example – step 5

Bayes' Theorem: $P(Y \mid X) = \dfrac{P(X \mid Y)P(Y)}{P(X)}$

Training Data

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Observe the conditional probabilities or likelihood the class non-mammals from 13 instances of training data

5 Likelihood: $P(X \mid Y = \text{non-mammals})$

Compute the probability of each attribute value given $Y =$ non-mammals. (There are 13 non-mammals.)

Attribute	Value in X (x_i)	Count with this value & $Y =$ non-mammals	Total non-mammals	$P(x_i \mid Y =$ non-mammals$)$
Give Birth	yes	1	13	$\frac{1}{13}$
Can Fly	no	10	13	$\frac{10}{13}$
Live in Water	yes	3	13	$\frac{3}{13}$
Have Legs	no	4	13	$\frac{4}{13}$

Multiply (Naïve Bayes assumption):

$$P(X \mid Y = \text{non-mammals}) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} \approx 0.0042$$

Counts come from the training table (13 non-mammals total).

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Another Example – step 6

Combine likelihood with priors

Bayes' Theorem: $P(Y \mid X) = \dfrac{P(X \mid Y)P(Y)}{P(X)}$

6 Combine with Priors (Unnormalized)

Multiply likelihoods by prior probabilities.

For $Y =$ mammals:

$$P(X \mid Y = \text{mammals})P(Y = \text{mammals}) = 0.06 \times \frac{7}{20} = 0.021$$

For $Y =$ non-mammals:

$$P(X \mid Y = \text{non-mammals})P(Y = \text{non-mammals})$$
$$= 0.0042 \times \frac{13}{20} = 0.0027$$

Which is larger?

$0.021 > 0.0027 \Rightarrow$ $Y =$ mammals is more likely.

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Another Example – step 6

Combine likelihood with priors

Bayes' Theorem: $P(Y \mid X) = \dfrac{P(X \mid Y)P(Y)}{P(X)}$

6 Combine with Priors (Unnormalized)

Multiply likelihoods by prior probabilities.

For $Y =$ mammals:

$$P(X \mid Y = \text{mammals})P(Y = \text{mammals}) = 0.06 \times \frac{7}{20} = 0.021$$

For $Y =$ non-mammals:

$$P(X \mid Y = \text{non-mammals})P(Y = \text{non-mammals})$$
$$= 0.0042 \times \frac{13}{20} = 0.0027$$

Which is larger?

$0.021 > 0.0027 \Rightarrow Y =$ mammals is more likely.

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Another Example – step 7,8

Bayes' Theorem: $P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$

7 Posterior Probabilities (Normalized)

Normalize so probabilities sum to 1.

Denominator:

$$P(X) = P(X \mid Y = \text{mammals})P(Y = \text{mammals}) + P(X \mid Y = \text{non-mammals})P(Y = \text{non-mammals})$$
$$= 0.021 + 0.0027 = 0.0237$$

Class (Y)	Unnormalized	Posterior Probability $P(Y \mid X)$
Y = mammals	0.021	$\frac{0.021}{0.0237} = 0.886$ (88.6%)
Y = non-mammals	0.0027	$\frac{0.0027}{0.0237} = 0.114$ (11.4%)
Total	0.0237	1.000 (100%)

8 Decision

The test instance X is classified as:

Y = MAMMALS

because $P(Y = \text{mammals} \mid X) = 0.886$ is greater than $P(Y = \text{non-mammals} \mid X) = 0.114$.

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

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How to Estimate Probabilities from Data

- For continuous attributes:
  - Discretization: Partition the range into bins:
    - one ordinal attribute per bin
    - violates independence assumption
  - Probability density estimation:
    - Assume attribute follows a normal distribution
    - Use data to estimate parameters of distribution (e.g., mean and standard deviation)
    - Once probability distribution is known, use it to estimate the conditional probability $P(X_i|Y)$

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Estimate Probabilities from Continuous Data by Probability density estimation

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Normal distribution:

$$P(X_i \mid Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(X_i-\mu_{ij})^2}{2\sigma_{ij}^2}}$$

  - One for each (X_i, Y_i) pair

- For (Income, Class=No):
  - If Class=No
    - sample mean = 110
    - sample variance = 2975

$$P(Income = 120 \mid No) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

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Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

What is the class label using Bayes Classifier?

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

Assume income follows normal distribution.

- *Class No: average=110, variance =2975*
- *Class yes: average = 90, variance =25*

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Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

For class No: average=110, variance =2975

For class yes: average = 90, variance =25

- Find: P(No|X) , P(Yes|X) which is greater? If 1st P is greater, class label =No. otherwise, label=Yes

We need to calculate P(X|No), P(No), P(X|Yes), P(Yes)

P(No)=7/10=0.7

P(Yes)=3/10=0.3

P(X|No)=P(Refund=No|No)*P(Marital status=married|No)*P(income=120K|No)

P(Refund=No|No)=4/7

P(Marital status=married|No)=4/7

P(income=120K|No) assume a normal distribution = (exponential (120-110) / 2*2975)/sqr(2pi*2975) = 0.0072

Thus: P(X|No)= = 4/7 * 4/7 * 0.0072 = 0.0024

P(X|Yes)= P(Refund=No|Yes)*P(Marital status=married| Yes)*P(income=120K| Yes) =0

- P(No|X)=P(X|No)P(No)~~/P(X)~~=0.0024*0.7 greater
- P(Yes|X)=P(X|Yes)P(Yes)~~/P(X)~~ = 0

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Issues with Naïve Bayes Classifier

- If one of the conditional probabilities is zero, then the entire expression becomes zero
- Need to use other estimates of conditional probabilities than simple fractions
- Probability estimation:

$$\text{original: } P(X_i = c|y) = \frac{n_c}{n}$$

$$\text{Laplace Estimate: } P(X_i = c|y) = \frac{n_c + 1}{n + v}$$

$$\text{m} - \text{estimate: } P(X_i = c|y) = \frac{n_c + mp}{n + m}$$

n: number of training instances belonging to class y

n_c: number of instances with $X_i = c$ and $Y = y$

v: total number of attribute values that X_i can take (e.g. v=2 with possible values: Yes, No)

p: initial estimate of ($P(X_i = c|y)$ known apriori

m: hyper-parameter for our confidence in p

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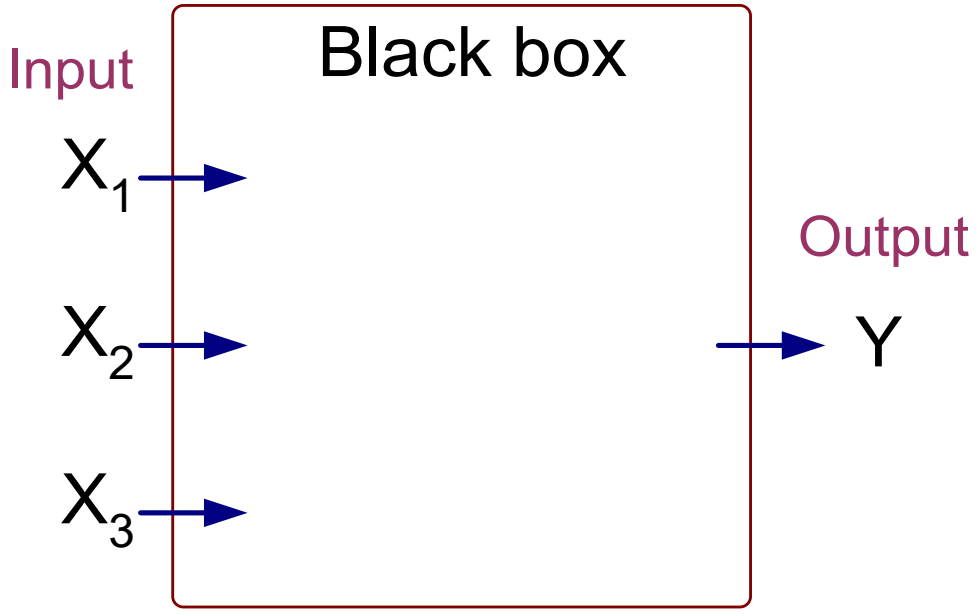
Naïve Bayes (Summary)

- Robust to isolated noise points
- Handle missing values by ignoring the instance during probability estimate calculations
- Robust to irrelevant attributes
- Redundant and correlated attributes will violate class conditional assumption
  - Use other techniques such as Bayesian Belief Networks (BBN)

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Artificial Neural Networks (ANN)

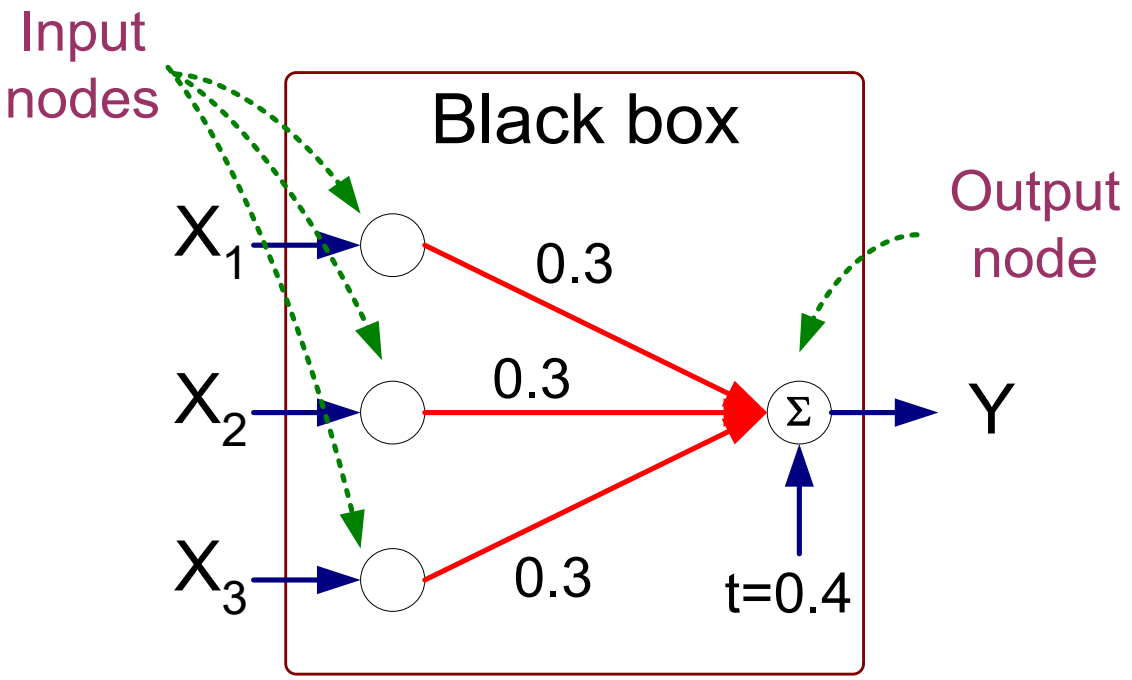
X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



Output Y is 1 if at least two of the three inputs are equal to 1.

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X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



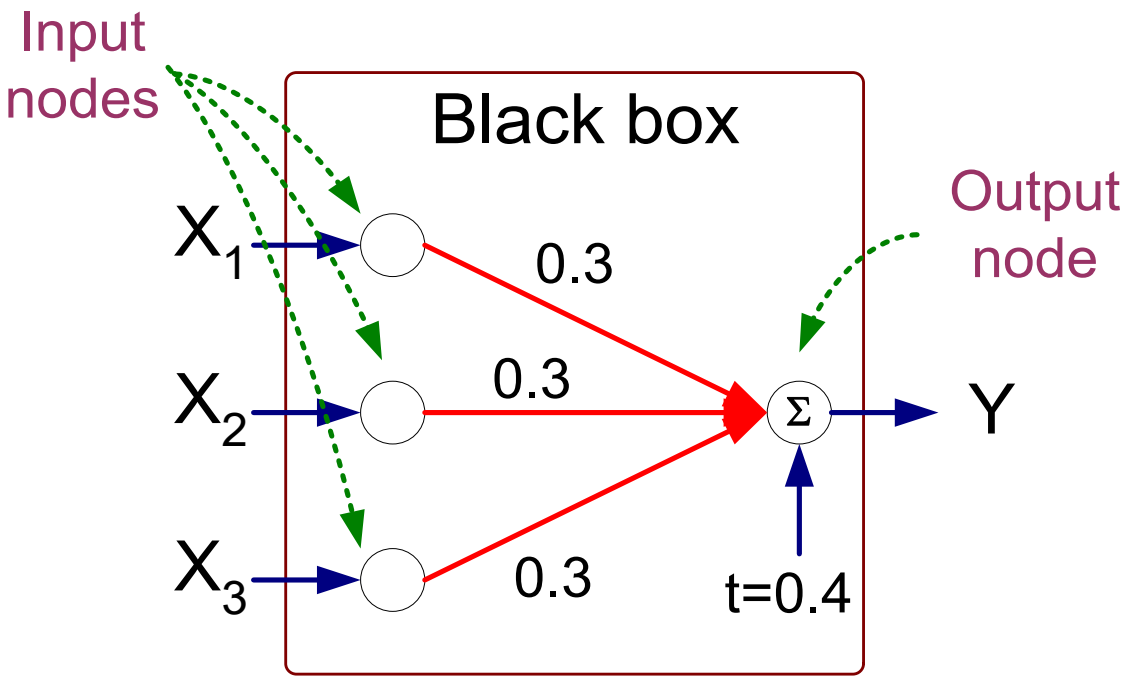
$$Y = sign(0.3X_1 + 0.3X_2 + 0.3X_3 - 0.4)$$

$$\text{where } sign(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

t is threshold = 0.4

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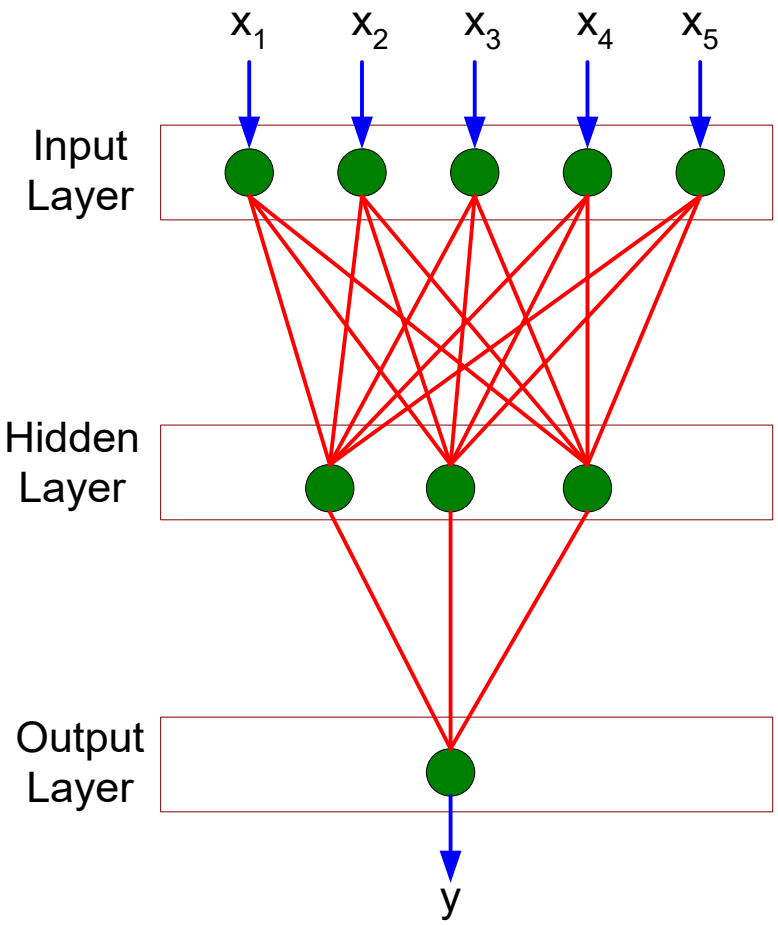
X_1	X_2	X_3	Y
1	0	0	-1
1	0	1	1
1	1	0	1
1	1	1	1
0	0	1	-1
0	1	0	-1
0	1	1	1
0	0	0	-1



- Model is an assembly of inter-connected nodes and weighted links
- Output node sums up each of its input value according to the weights of its links
- Compare output node against some threshold t

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General structure of Multi-layer NN



- More than one *hidden layer* of computing nodes
- Every node in a hidden layer operates on activations from preceding layer and transmits activations forward to nodes of next layer
- Also referred to as “feedforward neural networks”

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Algorithm for learning ANN

- Initialize the weights (w_0, w_1, …, w_d)
- Repeat
  - For each training example (x_i, y_i)
    - Compute $\hat{y}_i$
    - Update the weights:

$$w_j^{(k+1)} = w_j^{(k)} + \lambda\left(y_i - \hat{y}_i^{(k)}\right)x_{ij}$$

- Until stopping condition is met

k: iteration number; λ: learning rate

Symbol	Meaning
$w_j^{(k)}$	current weight for feature j
$w_j^{(k+1)}$	new updated weight
λ	learning rate
y_i	actual/true class
$\hat{y}_i$	predicted class
x_{ij}	value of feature j for training example i
k	iteration number

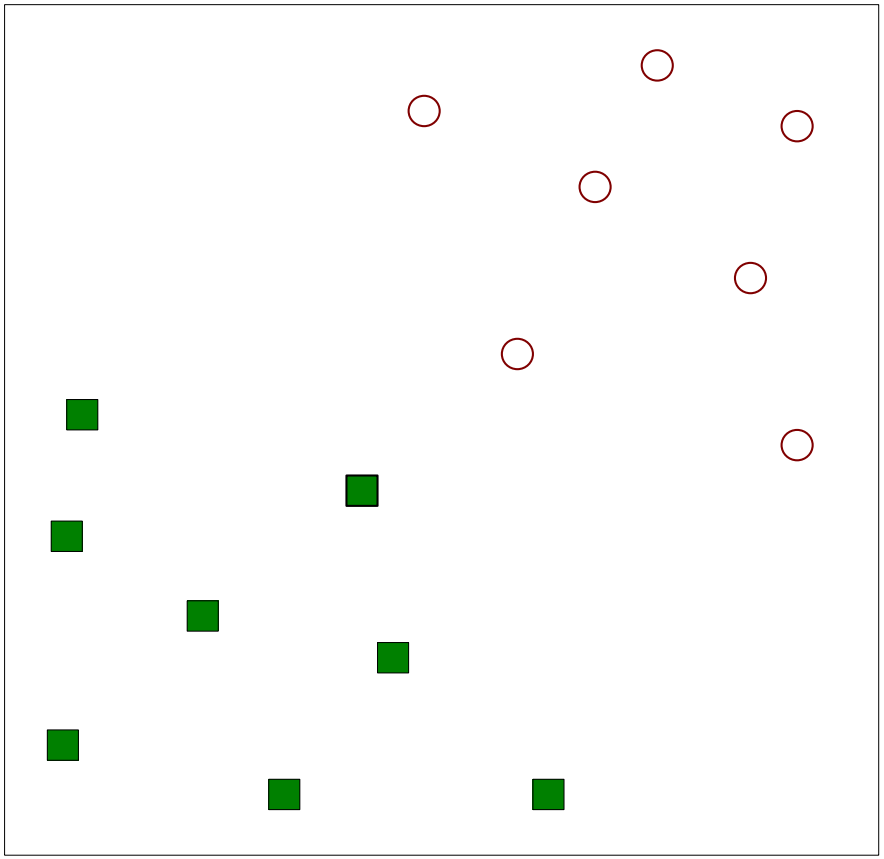
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Deep Learning Trends

- Training **deep** neural networks (more than 5-10 layers) could only be possible in recent times with:
  - Faster computing resources (GPU)
  - Larger labeled training sets
- Algorithmic Improvements in Deep Learning
  - Responsive activation functions (e.g., RELU)
  - Regularization (e.g., Dropout)
  - Supervised pre-training
  - Unsupervised pre-training (auto-encoders)
- Specialized ANN Architectures:
  - Convolutional Neural Networks (for image data)
  - Recurrent Neural Networks (for sequence data)
  - Residual Networks (with skip connections)
- Generative Models: Generative Adversarial Networks

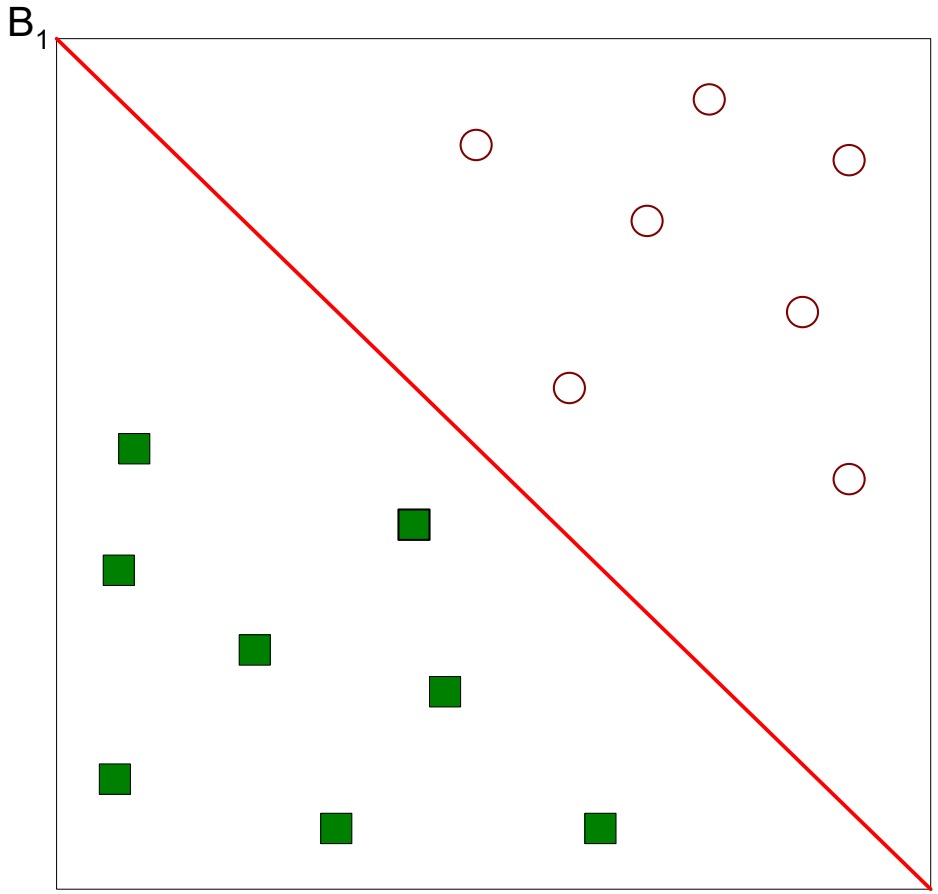
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Support Vector Machines



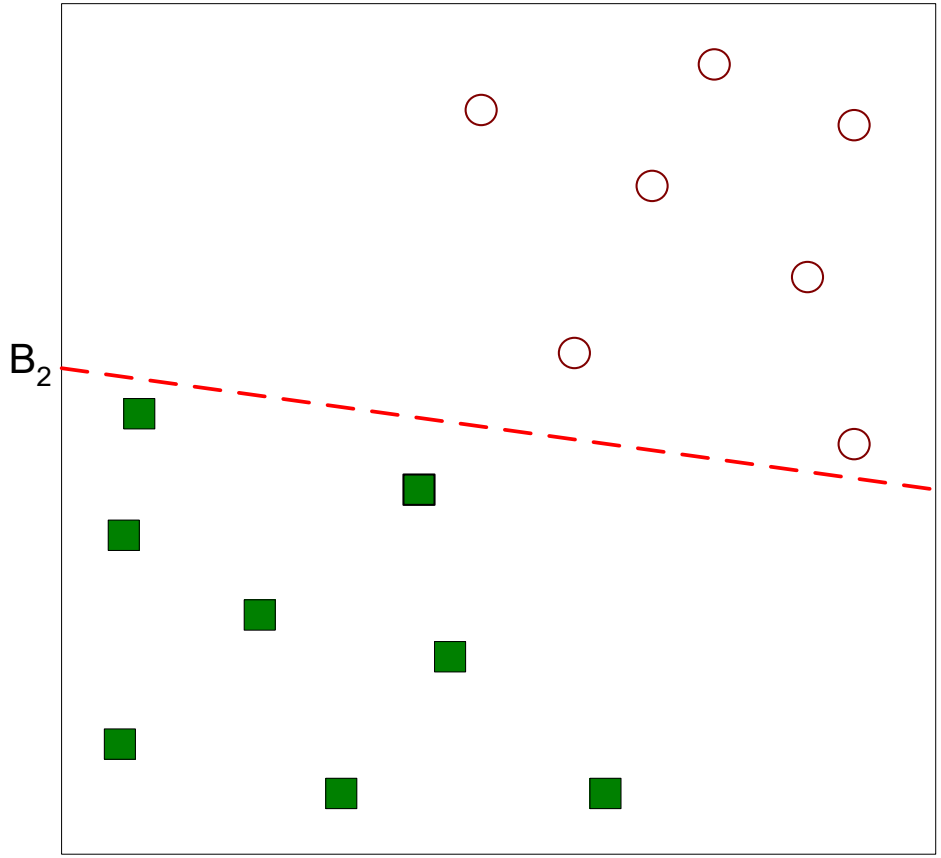
Find a linear hyperplane (decision boundary) that will separate the data

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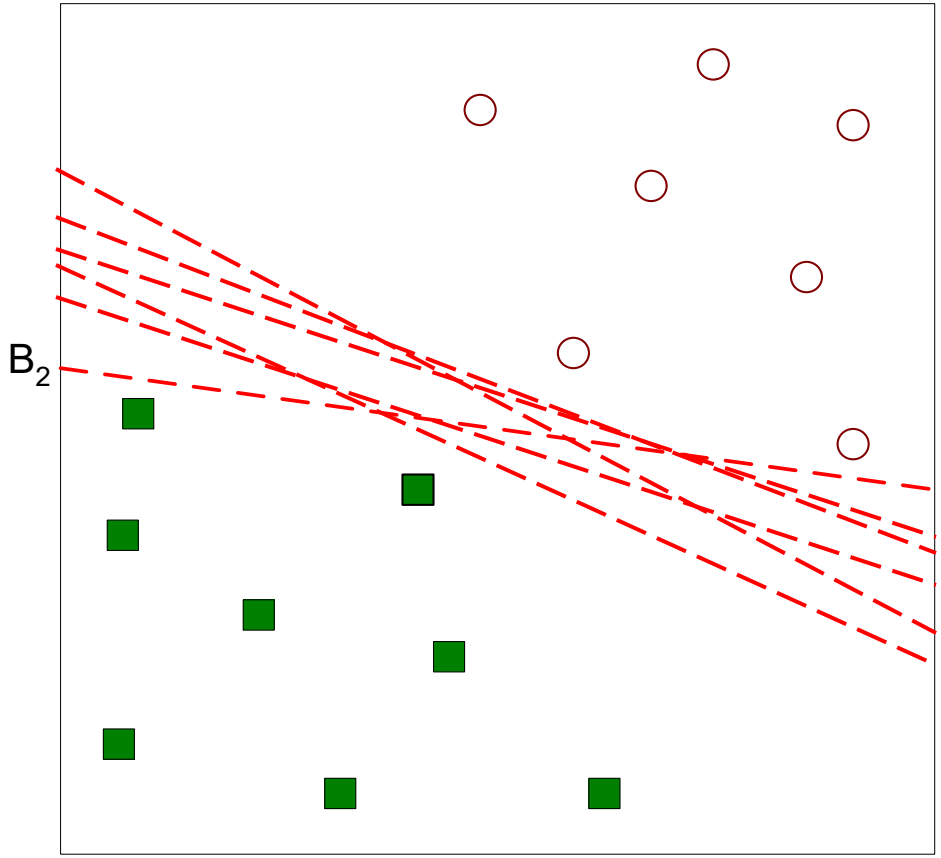
One Possible Solution

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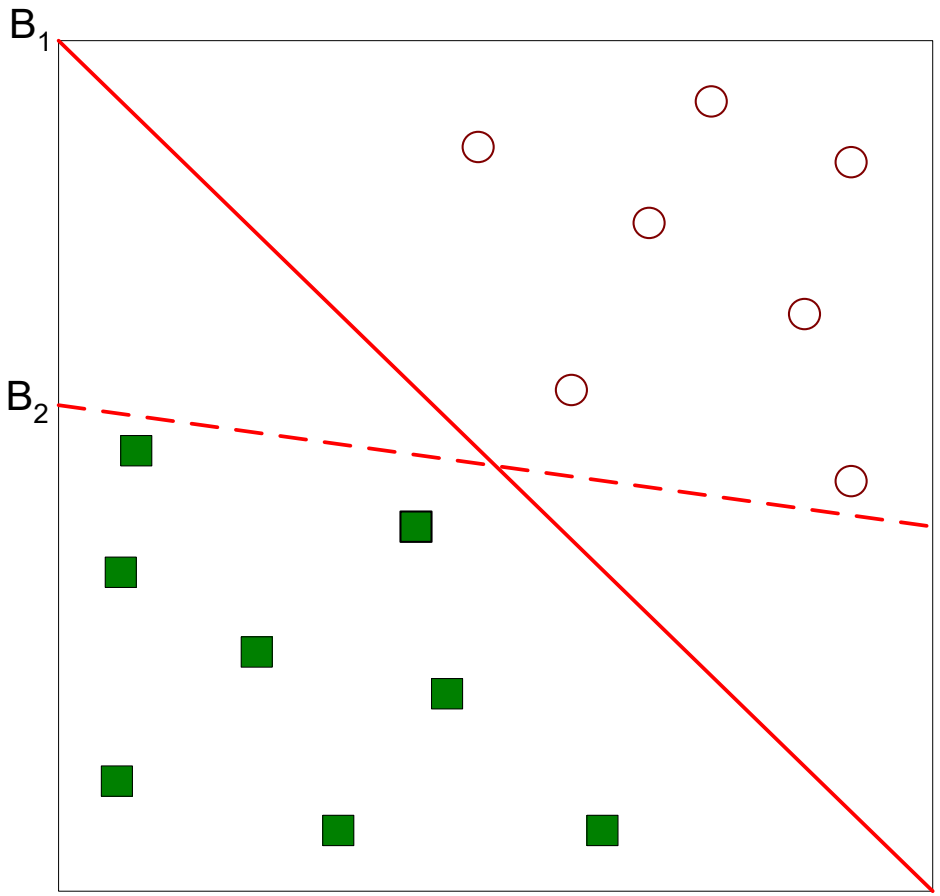
Another possible solution

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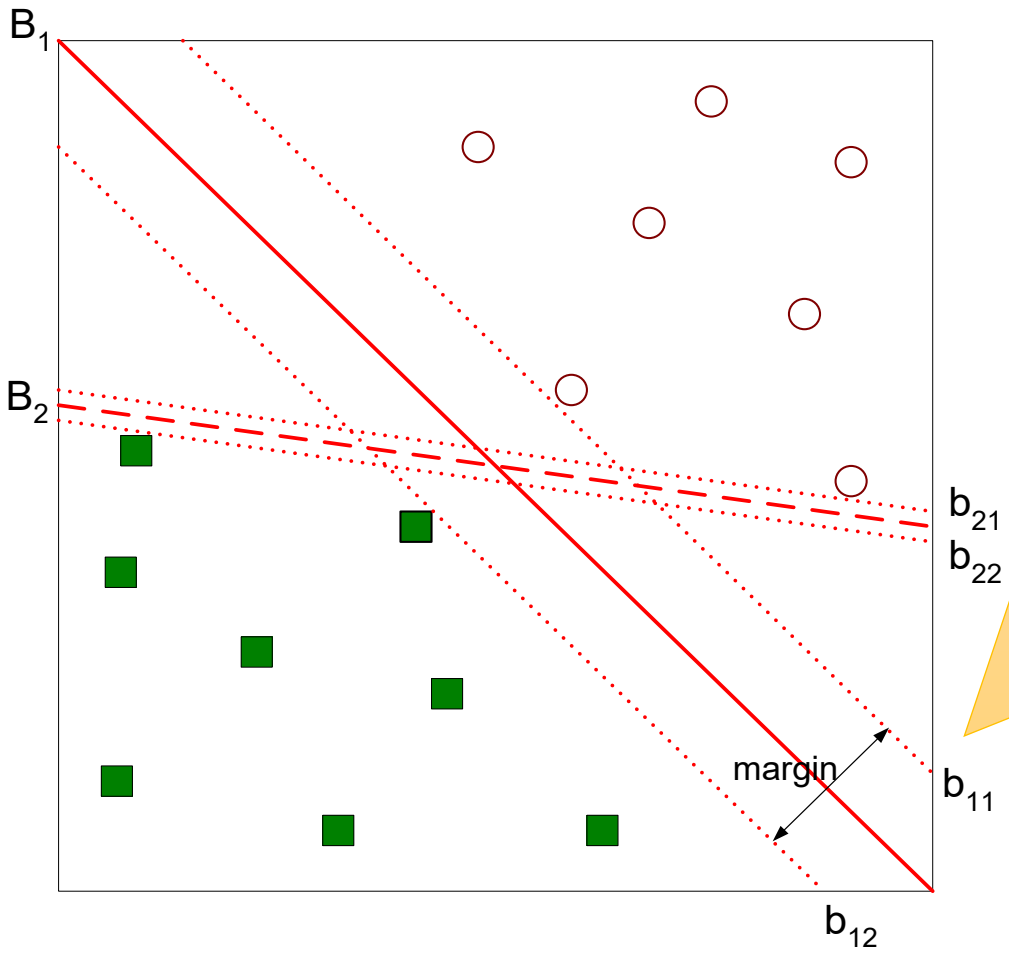
Other possible solutions

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- Which one is better? B1 or B2?
- How do you define better?

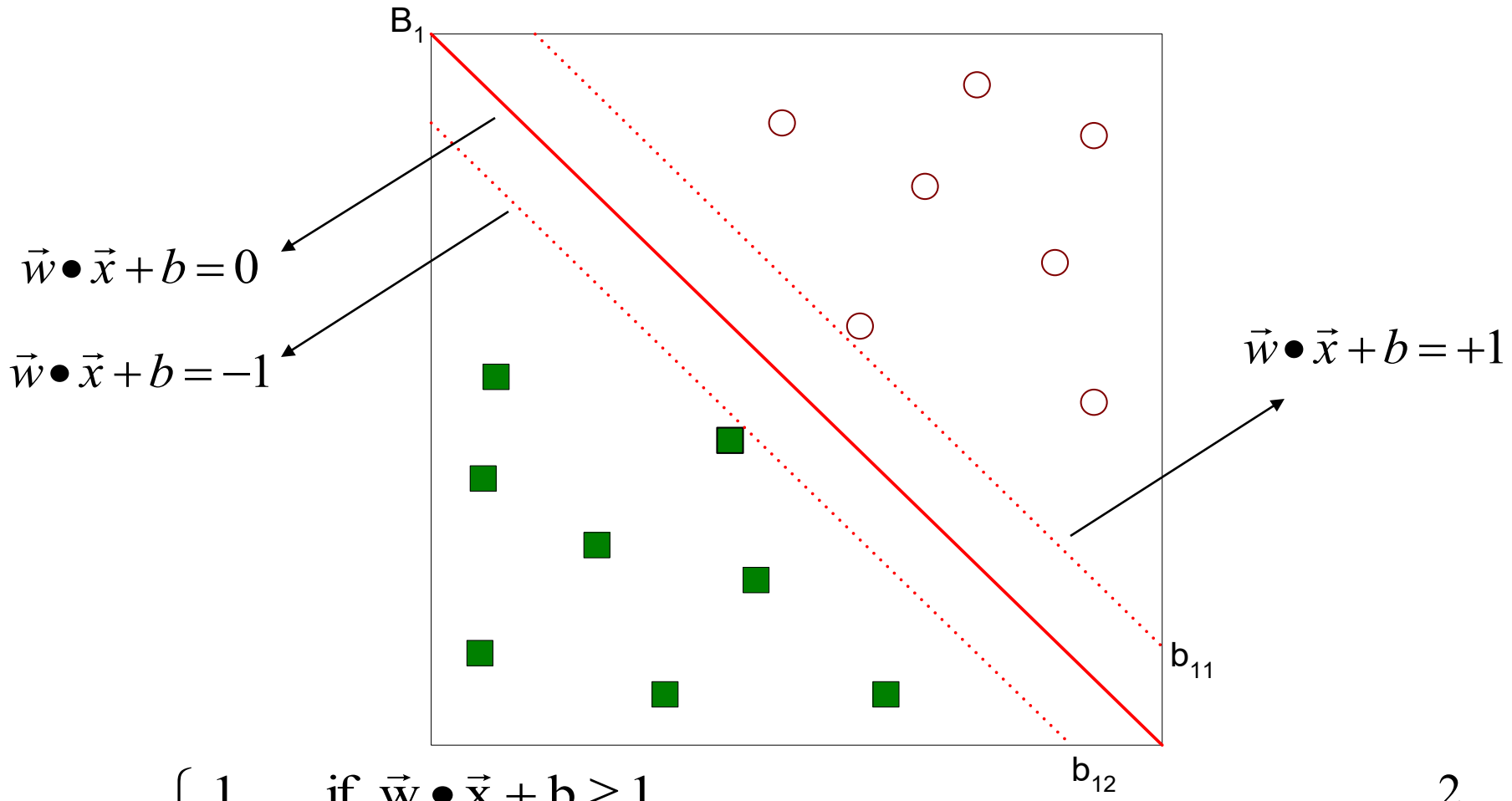
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Hyperplanes with larger margins tend to have better generalization performance than those with small margins

- Find hyperplane maximizes the margin => B1 is better than B2

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$$f(\vec{x}) = \begin{cases} 1 & \text{if } \vec{w} \bullet \vec{x} + b \geq 1 \\ -1 & \text{if } \vec{w} \bullet \vec{x} + b \leq -1 \end{cases}$$

$$\text{Margin} = \frac{2}{\| \vec{w} \|}$$

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- Objective is to maximize: $\text{Margin} = \frac{2}{\|\vec{w}\|}$

$$\|\mathbf{w}\| = \sqrt{w_1^2 + w_2^2 + \cdots + w_d^2}$$

length (magnitude) of the weight vector w

  - Which is equivalent to minimizing: $L(\vec{w}) = \frac{\|\vec{w}\|^2}{2}$
  - Subject to the following constraints:

$$y_i = \begin{cases} 1 & \text{if } \vec{\mathrm{w}} \bullet \vec{\mathrm{x}}_\mathrm{i} + \mathrm{b} \geq 1 \\ -1 & \text{if } \vec{\mathrm{w}} \bullet \vec{\mathrm{x}}_\mathrm{i} + \mathrm{b} \leq -1 \end{cases}$$

or $$y_i(\mathrm{w} \bullet \mathrm{x}_i + b) \geq 1, \qquad i = 1,2,\ldots,N$$

- This is a constrained optimization problem
  - Solve it using Lagrange multiplier method

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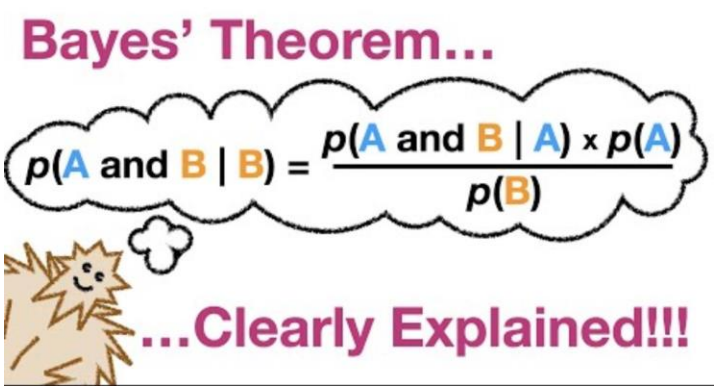
Characteristics of SVM

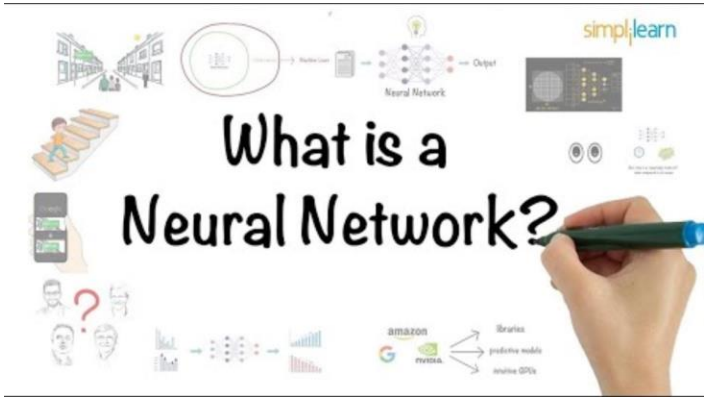
- The learning problem is formulated as a convex optimization problem
  - Efficient algorithms are available to find the global minima
  - Many of the other methods use greedy approaches and find locally optimal solutions
  - High computational complexity for building the model

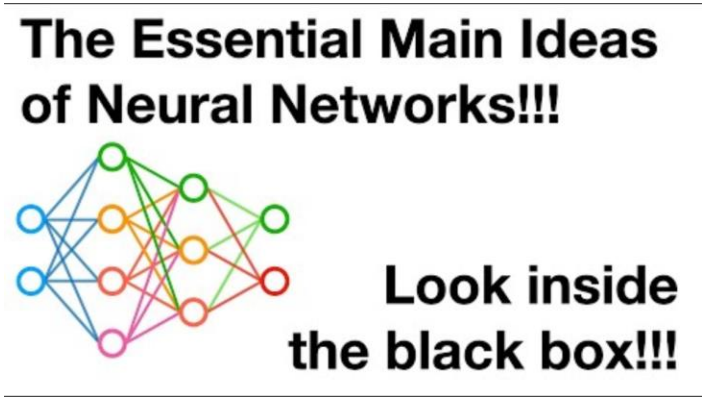
- Robust to noise
- Overfitting is handled by maximizing the margin of the decision boundary,
- SVM can handle irrelevant and redundant attributes better than many other techniques
- The user needs to provide the type of kernel function and cost function
- Difficult to handle missing values

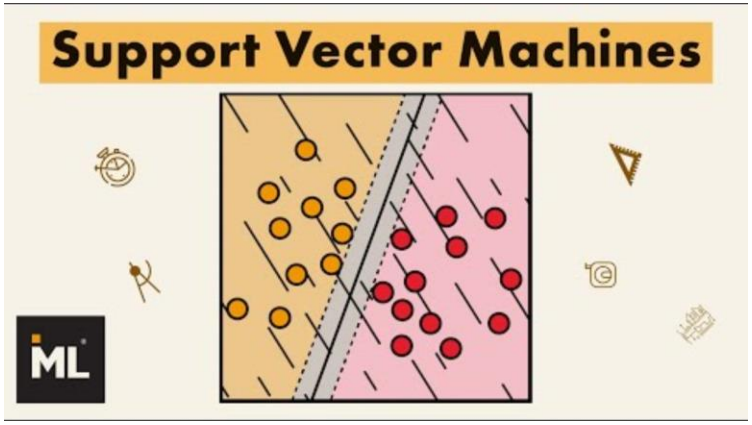
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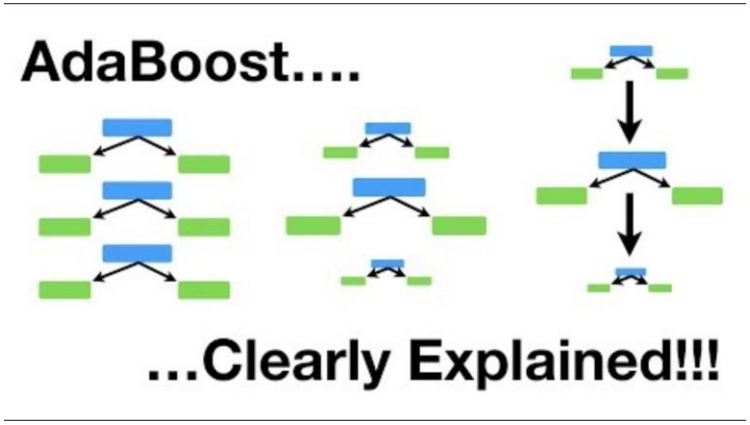
Suggested after-class readings (link in LMS)











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Summary

Bayes Classifier

- Uses probability to predict the most likely class.
- Main idea: calculate the posterior probability P(Y|X).
- Bayes theorem: P(Y|X) = P(X|Y) × P(Y) / P(X)
- Naïve Bayes simplifies the calculation by assuming attributes are conditionally independent.

Artificial Neural Networks (ANN)

- Learn the relationship between inputs and outputs using **weights**.
- Structure: Input Layer → Hidden Layer(s) → Output Layer
- Training repeatedly adjusts weights to reduce prediction error.
- Deep learning extends ANN using multiple hidden layers.

Support Vector Machines (SVM)

- Find a decision boundary that separates classes.
- Main idea: choose the boundary with the **maximum margin**.
- Decision boundary: $w \cdot x + b = 0$
- The closest observations are called **support vectors**.

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Thank you

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